INF 232

HCI & Health

1/20/2026

Overview of Health

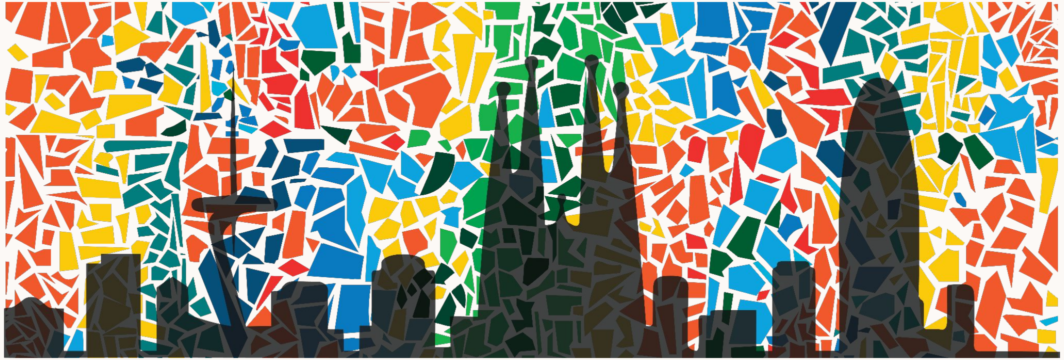
Core Topics in Health for CHI

- Health, wellness, and medicine, including:
  - Physical, mental, and emotional well-being
  - Clinical environments
  - Self-management and everyday wellness

Selecting a subcommittee. ACM CHI 2026. (2025, December 27). https://chi2026.acm.org/authors/papers/selecting-a-subcommittee/#health

Core Research Questions

- How do people interact with technology in healthcare contexts?
  - And what new understandings does the research contribute to HCI?
- How is it used in both formal and informal contexts (clinical vs. everyday use)?
- How does it impact stakeholders?
- How can health and wellness technologies be designed?



Selecting a subcommittee. ACM CHI 2026. (2025, December 27). https://chi2026.acm.org/authors/papers/selecting-a-subcommittee/#health

Disciplines Outside of HCI Informing Health Research

- Medicine & Clinical Science
- Psychology
- Public Health
- Sociology & Anthropology
- Data Science
- Engineering
- Ethics & Humanities

Similarities in Research Contributions Across Venues

CHI, JAMIA, AMIA all..

- Publish peer-reviewed health informatics research
- Value the advancement of health outcomes and its real world impact
- Use diverse methodologies and welcome interdisciplinary research

Differences in Research Contributions: Scope

CHI

- Health, wellness, medicine

- Physical, mental, emotional well-being

- Clinical environments, self-management, everyday wellness

https://chi2026.acm.org/authors/papers/selecting-a-subcommittee/#health

JAMIA

- Clinical care, clinical research

- Translational & implementation science, Imaging

- Education, consumer health, public health, policy

https://academic.oup.com/jamia/pages/General_Instructions?login=true

AMIA

Has <u>**three core areas**</u>:

- Translational bioinformatics

- Clinical Research Informatics

- Data Science/Artificial Intelligence

https://amia.org/education-events/amplify-informatics-conference/summit-proposals

AMIA Highlighted Themes

1. Translational Bioinformatics Using Multi-Modal Patient Data and AI
2. Novel Methods for Variant Detection and Interpretation from Omics Data
3. Emerging Best Practices and Operational Excellence in Scalable, Reproducible Clinical Research Informatics
4. Advances in Digital Health Technologies for Patient Engagement, Monitoring, and Research
5. Innovations in Health Data Science and Artificial Intelligence
6. Leveraging Large Language Models in Health Data Science
7. Generating and Applying Real-World Evidence through Informatics
8. Proactive and Predictive Machine Learning for Precision Biomedical Applications
9. Implementation Science and Deployment in Informatics
10. Seamless Integration of Clinical Research and Care Workflows

https://amia.org/education-events/amplify-informatics-conference/summit-proposals

Differences: Type of Contributions

CHI

- Empirical, theoretical, conceptual, methodological, design, & systems contributions

- Must have **novel HCI contribution** about people's interaction with technology or design of health/wellness tech

https://chi2026.acm.org/authors/papers/selecting-a-subcommittee/#health

JAMIA

- Describe innovative informatics research and systems that help to **advance** biomedical science & promote health

- Case reports, perspectives, reviews

https://academic.oup.com/jamia/pages/General_Instructions?login=true

AMIA

- Methodological innovations, original approaches

- Comprehensive evaluations that **propel** the biomedical research enterprise through informatics

https://amia.org/education-events/amplify-informatics-conference/summit-proposals

What is Health's history at CHI?

Health was under **“Featured Communities”**

Health was under **“Specific Application Areas”**

Health was under **“Health, Accessibility, and Aging”**

Health stands alone for the first time!!!! **“HEALTH”.**

(Bye, Accessibility and Aging, they are their own subcommittee now)

2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023 2024 2025

In 2010, **Madhu Reddy & Gillian Hayes** submitted a proposal in 2010 for a creation of new Health Community as a part of ACM SIGCHI and CHI conference, and the rest is a history!

How did the Descriptions on Health evolve over time?

2011 Health description under 'Featured Communities'

"Health is a fundamental aspect of human life…

Contributions to this community focus on a wide variety of issues related to health, both within and outside of a clinical setting…

Interactive systems for health, therefore, can include **assistive technologies, personal informatics and persuasive technologies, clinical systems, home healthcare devices and communication technologies, telemedicine, and much more."**

https://chi2011.acm.org/

2015 Health description under 'Specific Application Areas'

"This subcommittee will focus on papers that extend the design and understanding of applications for **specific application areas or domains of interest to the HCI community.**

Examples of potential user groups of interest include, but are not limited to: older adults, children, families, disabled people, people in developing countries, and people with perceptual, cognitive, or motor impairments.

Examples of application areas include, but are not limited to: education, **health,** home, sustainability, ict4d, security, privacy and creativity."

https://chi2015.acm.org/

How did the discussions on Health evolve over time?

2017 Health description under ‘Health, Accessibility, and Aging’

“This subcommittee is suitable for contributions to **independent and healthy living over a lifetime.** It combines the areas of (i) accessibility for people with disabilities**, (ii) health, wellness, and aging;** and, (iii) technology for and studies involving older adults.

Submissions to this subcommittee will be evaluated in part based on their inclusion of and potential impact on their target user groups and other stakeholders….”

https://chi2017.acm.org/

2019 Health description under ‘HEALTH’!

“This subcommittee is suitable for contributions related to **health, wellness, and medicine, including physical, mental, and emotional well-being, clinical environments, self-management, and everyday wellness.** …

This subcommittee welcomes all contributions related to **health, including empirical, theoretical, conceptual, methodological, design, and systems contributions….”**

https://chi2019.acm.org/

How did the discussions on Health evolve over time?

2020 Health description under 'HEALTH'!

New Disclaimer::

"Note that if your paper's topic is on "health of marginalized groups", it can potentially fit the description of Health and Specific Apps subcommittees. ....

If your contribution is about how health or interaction with the healthcare system was improved for any population, then submission should be to Health.

If your contribution is more about the marginalized community, then the submission should go to Specific Apps."

https://chi2020.acm.org/

2021 ~ Now
Health description under 'HEALTH'!

"..... The research problem can be grounded in **both formal and informal health and care contexts.** Submissions to this subcommittee will be evaluated in part based on **their inclusion of and potential impact on their stakeholders.**

Papers must have a clear and novel contribution to HCI in terms of **our understanding of people's interaction with technology in a healthcare context, or the design of health and wellness technologies.** For example, systematic reviews or usability studies associated with clinical trials must also have contributions for the HCI community."

https://chi2021.acm.org/

History of “Health” in CHI

How did the discussions on Health evolve over time?

2012

1. Healthcare + technology: putting patients first

2. Interfaces for health & wellbeing

3. Health + Design

4. Health & Children

2013

1. Communicating health

2. Food and Health

3. Mental Health (1st appearance)

2017

1. Mental health
2. Self-tracking Mental health
3. Self-monitored healthcare
4. Social computing and health
5. Enabling healthy behaviors
6. Family health
7. Health volunteers (health workers, public health)

2023

1. AI for health (1st appearance)
Health behavior change
2. Health data and tracking
3. Health and participatory design
4. Health and social media
5. Health over a distance
6. Managing health and care-giving
7. Mental health
8. Mental health and care work
9. Mental health and emotion

History of “Health” in CHI

How did the discussions on Health evolve over time?

2024

1. Digital healthcare and communication
2. Digital Wellbeing A & B
3. Health Ecosystems
4. Health and AI A & B & C
5. Health and Care practices
6. Healthcare Training
7. Healthy Aging
8. Health
9. Menstrual Tracking and Health
10. Mental Health A & B
11. Mental health and AI
12. Wellbeing and eating nutrition and weight
13. Wellbeing and mental health A & B & C

2025

1. Digital health and Wellbeing
2. Digital health for diverse needs
3. Eating and digital health
4. Health and expression support
5. Health and Wellbeing
6. LLM for Health
7. Lifetime digital health
8. Mental wellbeing
9. Mental and emotional wellbeing
10. Workplace interactions and wellbeing
11. Wellbeing and data tracking
12. Wellbeing and tracking

2026

What do you think where Heath is going now?

How has the focus of Health change?

2012: The “Clinical and Self-management” Era

Health Research’s Focus

- Novel sensing techniques
- Structured behavior change

Best papers/ Mention

- "Touché: Enhancing Touch Interaction on Humans, Screens, Liquids, and Everyday Objects" (Sato et al., 2012)
- "I did that! Measuring users' experience of agency in their own actions" (Coyle et al., 2012)

2025: The “Personal and AI-Integrated” Era

Health Research Focus:

- Personal health experience
- AI integrated

Best papers/ Mention:

- "Towards Hormone Health: An Autoethnography of Long-Term Holistic Tracking to Manage PCOS” (Kang et al., 2025)
- "(Re)discovering Sexual Pleasure after Cancer: Understanding the Design Space" (Offerman et al., 2025)
- “It Brought the Model to Life: Exploring the Embodiment of Multimodal I3Ms for People who are Blind" (Reinder et al., 2025)

Paper discussion

📝 Overview of Framing Papers

How to evaluate technologies for health behavior change in HCI research Klasnja et al., 2011

- States that behavior change requires “large scale, long term” studies that are infeasible for early-stage HCI technologies.
- Instead, focus should be tailored to specific behavior-change intervention strategies (e.g. self monitoring).
- We should understand how the design of technology affects real-world use by its target audience.

A Review of 25 years of CSCW Research in Healthcare: Contributions, Challenges, and Future Agendas Fitzpatrick & Ellingsen, 2012

- The paper maps the field's evolution (in CSCW) from early hospital-based systems to modern "pervasive" health, analyzing literature to review how we study healthcare technology.
- It establishes that healthcare is not just a technical problem but a deeply social and collaborative practice.
- The paper suggests that while the research is great at describing small-scale work, it is failing to influence large-scale policy.

Core Sub-topics (What we study)

➢ **The papers suggest that the Health subcommittee's scope is broad, covering both personal and professional domains.**

- Personal Health & Lifestyle: Self-care and everyday wellness
- Chronic Disease Management: (e.g. coronary heart disease, diabetes)
- Health behavior change: technologies that support & evaluate changes

- Clinical & Specialized Workplace coordination: Healthcare work, Operation Rooms, ICUs, and Radiology department
- Clinical documentation: Electronic Health Records (EHR) and paper-based artifacts
- Distributed & collaborative care : Multidisciplinary team meetings
- Expanding Care Contexts

Methodological Approaches (How we study it)

➢ **Health research seems to move away from purely “clinical outcomes” to more “HCI-centered evaluations”**

- Evaluating Proximal outcomes
- Use field evaluations for early stages of development
- Run weeks-long, small studies with significant qualitative component

- Ethnographic and qualitative methods (workplace studies)
- Small-scale prototyping - short-term, localized prototypes/ Design-led prototypes
- Multi-site studies: to improve generalizability
- Bottom-up perspectives: human-centeredness; clinician’s perspectives

Theoretical Perspectives (The way we view them)

➢ **“Health” research in HCI uses theories to explain why technology works (or doesn’t)**

- Psychology:
  - Behavior change: long term, relapse-prone process
  - Priming: subtle cues that keep health goals in mind and guide behavior
  - Social influence: behavior shaped by social context & other people

- Sociology:
  - Articulation work (Sociology of Work : work to get the work done)
  - Socio-Technical Systems - viewing healthcare as a complex mix of social, cultural and political factors.
  - Information systems: Information Infrastructure
  - Mobility Work

1/22/2026

Discussion of Health

Before we discuss the papers

Why did we choose these papers?

- Selection of high-impact, award-recognized research (best papers) from the most recent 2024-25 CHI conferences.
- Trending avenues in Personal Health
  - **Holistic hormone health:** moving beyond traditional tracking towards long-term, systematic personal health.
  - **AI-augmented clinical care:** exploring the intersection of LLM and specialized healthcare interventions (smoking cessation)
- Refreshing Research contribution!
  - **Methods:** autoethnographies… And LLM evaluation pipelines….!!!!!

Before we discuss the papers

Titles coincidence?

Towards Hormone Health: An Autoethnography of Long-Term Holistic Tracking to Manage PCOS. Kang et al., 2025

Towards AI-Driven Healthcare: Systematic Optimization, Linguistic Analysis, and Clinicians' Evaluation of Large Language Models for Smoking Cessation Interventions. Calle et al., 2025

What does the word "Towards" in the title of academic research indicate?

- Exploratory, foundational or directional ...

1 **Towards Hormone Health: An Autoethnography of Long-Term Holistic Tracking to Manage PCOS.** Kang et al., 2025

Quick Overview of the paper

- **Goal:**
  - To explore the challenges of managing PCOS, which is a complex and highly individualized hormonal condition that often lacks reasonable technical support
  - Aims to shift the focus from simple fertility/menstrual monitoring to *hormonal health*
- **Methods:**
  - **A 10-month autoethnography** to track the first author's own holistic data (55 distinct variables such as physiological symptoms, emotional states, food intake, and other environmental factors…)
  - The first-person approach was chosen to capture the granular, daily cognitive loads that most 1-on-1 interviews tend to miss.

1 **Towards Hormone Health: An Autoethnography of Long-Term Holistic Tracking to Manage PCOS.** Kang et al., 2025

Quick Overview of the paper

- **Finding:**
  - Burden: The study highlights the burden involved in interpreting ambiguous health data and making lifestyle adjustment without clear guidance from existing technology.
  - Challenges: 1) Medical 2) Technical 3) Spatial 4) Temporal 5) Socio-cultural
- **Discussions:**
  - The paper proposes a shift towards "holistic tracking" in HCI, suggesting design implications that support the integration of subjective experiences with objective data to better manage long-term fluctuating conditions.
  - Design implications: future health tools should support "flexible data consolidation" and provide context-aware scaffolding

Paper discussion

1 **Towards Hormone Health: An Autoethnography of Long-Term Holistic Tracking to Manage PCOS.** Kang et al., 2025

Significance: why study Hormone Health?

Argument

- The paper moves beyond standard “fertility” or “menstrual” tracking to argue that hormonal health (PCOS) is a systemic, long-term, unique physiological state that existing episodic tracking fails to capture
- PCOS affects 11-13% of women, and presents with highly variable symptoms.

The Gap

- The author highlights the labor and frustration of managing chronic, fluctuating symptoms that don’t fit into neat data visualization.
- Current tools fail to support the personalization needed for this disorder.

1

Towards Hormone Health: An Autoethnography of Long-Term Holistic Tracking to Manage PCOS. Kang et al., 2025

Interdisciplinary Inquiry

HCI Core

- Personal Informatics and Self-tracking literature

Outside HCI

- Endocrinology (medical study of hormones) and Feminist Theory (sociology)

Expansion

- The paper expands by introducing **autoethnography** as a way to validate subjective lived experience as “hard data” for design…

Paper discussion

1

Towards Hormone Health: An Autoethnography of Long-Term Holistic Tracking to Manage PCOS. Kang et al., 2025

HCI Research Contributions

Contribution Type:

- Empirical, Methodological

Why Appropriate:

- The autoethnographic approach reveals not only physiological data but also emotional and cognitive load as a lived experience individual that other tracking methods might miss.

Takeaway:

- The paper challenges “reductionist” data tracking and introduces “holistic” approach.
- Future health tools should support “flexible data consolidation”

2

Towards AI-Driven Healthcare: Systematic Optimization, Linguistic Analysis, and Clinicians' Evaluation of Large Language Models for Smoking Cessation Interventions. Calle et al., 2025

Quick Overview of the paper

- **Goal:**
  - (1) Examine how to optimize LLMs to mimic human expert writing for smoking cessation messages.
  - (2) Evaluate whether LLM-generated smoking cessation messages meet clinical standards
- **Methods:**
  - Three total studies: prompt engineering, decoding optimization, and expert review
  - Message evaluation using automated metrics, linguistic analysis, and TTS (Tobacco Treatment Specialists) expert ratings

2

Towards AI-Driven Healthcare: Systematic Optimization, Linguistic Analysis, and Clinicians' Evaluation of Large Language Models for Smoking Cessation Interventions. Calle et al., 2025

Quick Overview of the paper

- **Findings:**
  - Simple, general prompts outperformed detailed instructions in generating expert-like messages.
  - States that larger LLMs (e.g. ChatGPT) **can** effectively emulate expert writing to generate well-written, accurate, and persuasive messages
    - ChatGPT exceeded human performance (71% vs 48% approval rating), but **~30%** expert rejection persists
- **Discussion**:
  - LLMs can augment smoking cessation interventions when evaluated through a combined framework of automated metrics, linguistic analysis, and expert review.

2

Towards AI-Driven Healthcare: Systematic Optimization, Linguistic Analysis, and Clinicians' Evaluation of Large Language Models for Smoking Cessation Interventions. Calle et al., 2025

Significance: why it's worthy of study?

Argument

- Smoking cessation interventions rely on large volumes of expert-written messages, making content creation labor-intensive and difficult to scale.
- LLMs present a promising alternative for automated health message generation, but their clinical safety and validity remain uncertain.

The Gap

- Automated metrics and non-expert evaluations cannot reliably assess safety, credibility, or treatment alignment.

2

Towards AI-Driven Healthcare: Systematic Optimization, Linguistic Analysis, and Clinicians' Evaluation of Large Language Models for Smoking Cessation Interventions. Calle et al., 2025

Interdisciplinary Inquiry

HCI Core

- Human-centered AI & digital health interventions

Outside HCI

- Natural Language Processing (NLP)
- Behavioral Health

Expansion

- The paper advances prior work by translating clinical standards for smoking cessation into an actionable evaluation and optimization process for LLMs.

2

Towards AI-Driven Healthcare: Systematic Optimization, Linguistic Analysis, and Clinicians' Evaluation of Large Language Models for Smoking Cessation Interventions. Calle et al., 2025

HCI Research Contributions

Contribution Type:

- Empirical, Methodological

Why Appropriate:

- Empirical: the authors test prompts, decoding strategies, and models using quantitative and expert evaluations.

- Methodological: the authors proposes a new evaluation framework that integrates automated metrics, linguistic attributes, and expert evaluations to assess LLMs in healthcare context

2

Towards AI-Driven Healthcare: Systematic Optimization, Linguistic Analysis, and Clinicians' Evaluation of Large Language Models for Smoking Cessation Interventions. Calle et al., 2025

HCI Research Contributions

Takeaway:

- Results indicate that larger LLMs can generate expert-like smoking cessation messages and effectively support clinical intervention workflows.
    - **However:** The study emphasizes that LLM-based automated message generation must incorporate model optimization, message filtering, comprehensive evaluation, and expert review to ensure safety, quality, and clinical effectiveness.

Questions?